

x86 CPU Governor Modeling Pipeline

MLP students and a student-warmed dual-frequency DDQN

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Problem and data

Teacher–student modeling

- Configuration branch

- Frequency branch

Student-warmed dual-head agent

ARM-only modeled-power results

Results and limitations

Objective

Use offline x86 measurements to learn a policy that selects:

1. an Ondemand governor configuration;
2. a legal CPU-frequency bin.

Work completed

- ▶ Configuration teacher and MLP student
- ▶ Frequency teacher and MLP student
- ▶ Student-only DDQN warming surface
- ▶ LOSO evaluation and latency measurement

Item	Value
Raw x86 runs	4,979
Utilization states	10
Sampling rates	10
Up-threshold values	5
Configurations per state	50
Aggregated state-action rows	500

Action grid

Sampling rate:

10,000, 20,000, ..., 100,000 μs

Up-threshold:

75, 80, 85, 90, 95%

Reference Ondemand action: 10,000 μs , 95%.

1. Compute EDP for each raw execution:

$$\text{EDP} = P_{\text{core}} \times \text{MRT}^2.$$

2. Group by targeted utilization, sampling rate, and threshold.
3. Retain median EDP, measured utilization, and observed frequency.

State

$$\mathbf{s} = [\rho, \log_{10}(1 - \rho)]$$

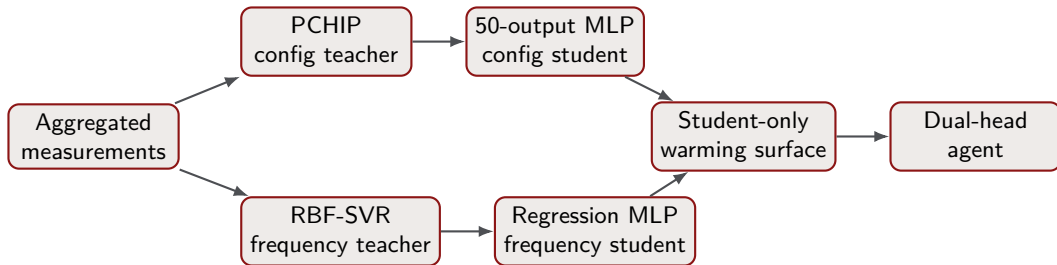
ρ is measured CPU utilization under the reference Ondemand configuration. The logarithmic term increases resolution near saturation.

Combined action

$$\mathbf{a} = (c, b)$$

$c \in \{0, \dots, 49\}$ is a sampling-rate/threshold configuration.

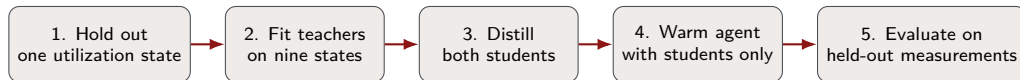
$b \in \{0, \dots, 31\}$ is a legal frequency bin.



Knowledge flow

The agent does not query the teachers during warming. Teacher knowledge reaches it indirectly through the two students.

Leave-one-state-out evaluation protocol



- ▶ Ten folds: each state is held out once with all 50 of its configurations.
- ▶ Three agent seeds per fold: 7, 42, and 123.
- ▶ Total: 30 committed agent decisions.
- ▶ Endpoint folds require extrapolation beyond the training-state range.

For each configuration c , the teacher models

$$y_c(\rho) = \log_{10}(\text{EDP}_c(\rho)) - \log_{10}(\text{EDP}_{\text{OD}}(\rho)) = \log_{10}\left(\frac{\text{EDP}_c(\rho)}{\text{EDP}_{\text{OD}}(\rho)}\right).$$

- ▶ Input to each curve: $\log_{10}(1 - \rho)$.
- ▶ Interpolator: shape-preserving PCHIP.
- ▶ Decision: $c^* = \arg \min_c y_c(\rho)$.
- ▶ A negative score means lower EDP than the Ondemand reference.

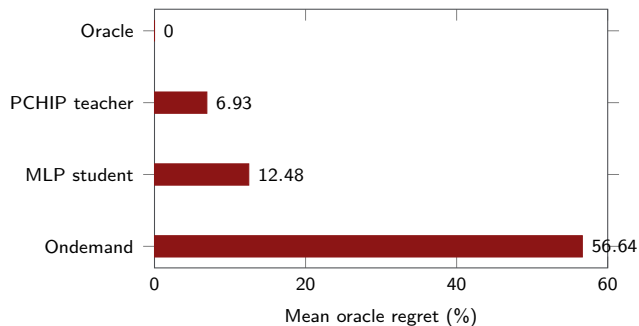
Why the Ondemand baseline does not change the optimum

At a fixed state, the same $\log_{10}(\text{EDP}_{\text{OD}})$ is subtracted from every configuration; therefore the ranking is unchanged.



7,602 trainable parameters

- ▶ 256 synthetic ρ points per fold
- ▶ Target: $q_c = -y_c$
- ▶ Standardize scores across 50 actions at each ρ
- ▶ Adam, learning rate 0.002
- ▶ 1,400 epochs; batch size 128
- ▶ Greedy decision: $\arg \max_c q_c$



Metric	Student
Mean regret	12.48%
Median regret	2.36%
Worst regret	87.33%
Exact oracle	40%
Within 5% of oracle	70%
Beats Ondemand	100%
Mean EDP vs Ondemand	-18.92%

Input features

$$[\rho, \log_{10}(1 - \rho), sr_{norm}, thr_{norm}]$$

The configuration metadata conditions the frequency estimate on the selected governor action.

Teacher model

- ▶ StandardScaler
- ▶ RBF-SVR
- ▶ $C = 100$
- ▶ $\gamma = 0.1$
- ▶ $\epsilon = 0.005$

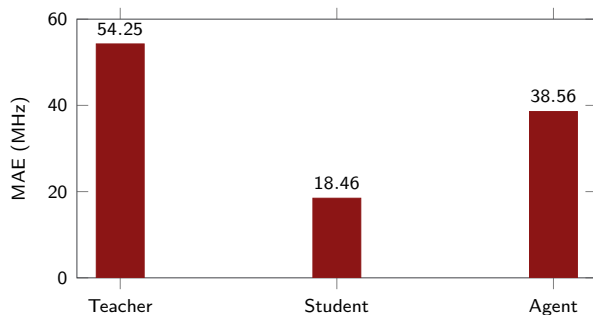
Target normalization:

$$f_{norm} = \frac{f - 800 \text{ MHz}}{2,100 - 800 \text{ MHz}}.$$



1,249 trainable parameters

- ▶ 64 synthetic ρ points
- ▶ 50 configurations per point
- ▶ 3,200 distillation examples
- ▶ Adam, learning rate 0.005
- ▶ 900 epochs; batch size 512
- ▶ Sigmoid keeps output in $[0, 1]$



Metric	Teacher	Student
MAE (MHz)	54.25	18.46
RMSE (MHz)	119.78	36.70
R^2	0.1957	0.9245
Illegal outputs	0	0

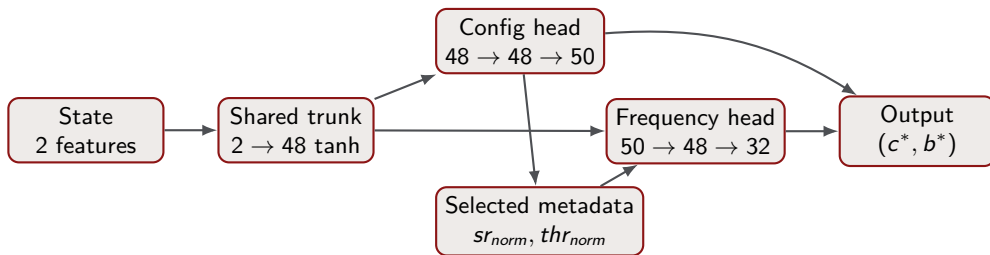
Important: the frequency branch imitates observed frequency; it does not directly optimize frequency for EDP.

The two students generate a dense synthetic dataset used by the agent:

Array	Shape	Meaning
S	96×2	State features
Q	96×50	Standardized configuration-student scores
F	96×50	Normalized frequency-student targets

Implementation property

The warming function receives the configuration student, frequency student, and action metadata. Teacher objects are absent from its interface and prediction path.



- ▶ Total parameters: 8,962.
- ▶ Factorization uses 50 configuration outputs and 32 frequency outputs instead of $50 \times 32 = 1,600$ joint outputs.

Frequency bins

- ▶ Range: 800–2,100 MHz
- ▶ Number of bins: 32
- ▶ Maximum nearest-bin gap: 20.3125 MHz

Immediate rewards

$$R_f = - \left(b_{norm} - \hat{f}_{student} \right)^2$$
$$R_c = Q_c + \lambda \max_b R_f, \quad \lambda = 2.$$

Parameter	Value	Parameter	Value
Training steps	2,000	Batch size	128
Learning rate	0.003	Discount γ	0
Initial ϵ	1.0	Final ϵ	0.05
Target update τ	0.05	Seeds	7, 42, 123

Training data

Exploration and replay occur on the synthetic student surface, not through interaction with a live CPU.

Effect of setting $\gamma = 0$

The standard DDQN target is

$$y = r + \gamma Q_{\text{target}}\left(s', \arg \max_a Q_{\text{online}}(s', a)\right).$$

With $\gamma = 0$:

$$y = r.$$

Still present

Replay batches, epsilon exploration, on-line/target networks, and two Q-value heads.

No longer present

Future-return bootstrapping, delayed credit assignment, and sequential planning.

The resulting learner is a DDQN-shaped immediate-reward contextual-bandit policy warmer.

Metric	Result
Mean oracle regret	12.48%
Median oracle regret	2.36%
Worst oracle regret	87.33%
Exact oracle choice	40%
Within 5% of oracle	70%
Beats Ondemand	30/30 decisions
Mean EDP relative to Ondemand	−18.92%
Mean seed standard deviation of regret	0.00
Frequency MAE vs measured value	38.56 MHz
Illegal frequency outputs	0

Result: the agent reproduces the configuration-student policy exactly across states and seeds; it does not improve that policy.

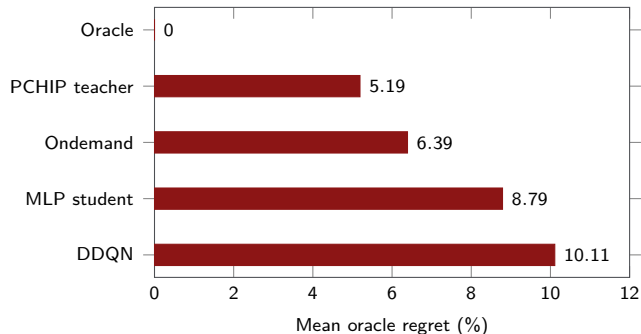
Model	Median (ms)	p95 (ms)	p99 (ms)
Configuration MLP	0.0147	0.0495	0.1024
Frequency MLP*	0.0327	0.0657	0.1089
Dual-head agent	0.0265	0.0547	0.0830

What changed from the x86 deck

- ▶ Train and evaluate on ARM only.
- ▶ EDP uses modeled CORE power, not measured CORE power.
- ▶ Both students are MLPs: configuration 64×64 and frequency 32×32 .
- ▶ DDQN warming uses only student outputs.

Evaluation protocol

- ▶ 5,000 ARM runs aggregated into 500 state-action rows.
- ▶ 10 held-out utilization states.
- ▶ 50 measured governor actions per state.
- ▶ 3 agent seeds: 7, 42, and 123.



Policy	Mean	Median	Worst
PCHIP teacher	5.19%	0.53%	24.57%
Ondemand	6.39%	0.91%	56.77%
MLP student	8.79%	0.84%	56.82%
DDQN agent	10.11%	0.85%	69.83%

Takeaway: on ARM, the student-warmed DDQN is stable but does not beat Ondemand on mean oracle regret.

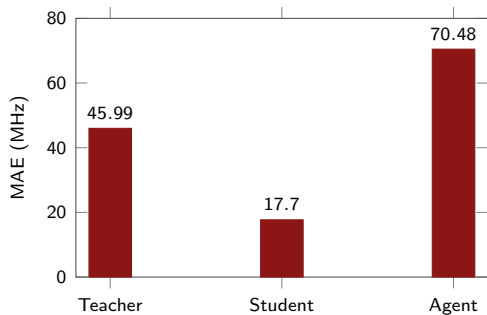
Policy	Exact oracle	Within 5%	Beats OD
PCHIP teacher	40%	80%	40%
Ondemand	40%	90%	0%
MLP student	40%	80%	20%
DDQN agent	30%	80%	23.3%

Agent deltas

- ▶ Mean regret is 3.72 percentage points worse than Ondemand.
- ▶ Mean regret is 1.32 percentage points worse than the configuration MLP.
- ▶ Seed standard deviation of mean regret is only 0.11 percentage points.

Interpretation

The ARM agent mostly inherits the student surface, but the DDQN warming step slightly degrades the configuration choice instead of improving it.



Model	Median	p95	p99
Config MLP	0.0137	0.0411	0.2118
Frequency MLP	0.0310	0.1029	0.3691
DDQN agent	0.0254	0.0428	0.0999

Agent frequency-bin MAE: 64.60 MHz vs student target and 70.48 MHz vs measured frequency.
Legal-frequency violations: 0%.

1. The ARM configuration teacher is best among learned policies: 5.19% mean oracle regret.
2. The MLP configuration student is usable but worse than teacher and Ondemand: 8.79% mean regret.
3. The student-warmed DDQN reaches 10.11% mean regret, with low seed variance but worse mean regret than Ondemand.
4. The frequency MLP is accurate as an imitation model: 17.70 MHz held-out MAE.
5. The agent is deployment-lightweight: 0.0254 ms median latency and 0% illegal frequency outputs.

1. The configuration student obtains 12.48% mean regret and beats Ondemand at every state.
2. The frequency student reduces MAE from 54.25 MHz to 18.46 MHz relative to the SVR teacher evaluation.
3. The dual-head agent preserves the student configuration policy and outputs only legal frequency bins.
4. Reward scaling and $\gamma = 0$ explain why the agent behaves primarily as a stable student-policy reproducer.